Soknarith (Soki) Sem

(401) 952–6819 — seminaia401@gmail.com — Seminaia — soknarith-sem— Worcester, MA

Summary

Materials engineer with experience in process optimization, manufacturing support, material characterization, and computational analysis across research and industrial environments. Background includes cGMP operations, statistical process control, root cause investigation, SOP development, and atomistic simulation. Brings a combined process and materials perspective to improving manufacturing efficiency, process reliability, and product quality.

Work Experience

Research Assistant & Graduate Teaching Assistant — **May 2024 to Present**
Worcester Polytechnic Institute - Worcester, MA

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling to evaluate material stability and performance.
- Analyzed structure–property–process relationships across temperature and chemical potential regimes to understand defect behavior relevant to semiconductor manufacturing.
- Assisted in teaching Metal Additive Manufacturing, Introduction to Materials Science and Engineering, Advanced Thermodynamics, and Applied Analytical Methods by grading assignments and leading homework help sessions.

Chemical Process Engineer — **May 2023 to May 2024**
Knowles Precision Devices - New Bedford, MA

- Led R&D-to-production transition for high-voltage capacitor dielectric fluids, supporting scale-up and manufacturing readiness.
- Optimized manufacturing layout and process flow using LEAN principles and statistical process controls, reducing production costs by 15%.
- Authored SOPs and work instructions for material handling and process scale-up, for training technicians and operators.
- Performed root cause analysis on material-related failures, including deviation investigation and CAPA implementation.

R & D Technician — **Mar 2023 to May 2023**
Factorial Energy - Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments to support researchers in evaluating battery performance under various conditions.
- Reduced material waste by 20% through inventory system redesign.

Downstream Purification Engineer — **Jul 2022 to Mar 2023**
Eurofins Scientific - Framingham, MA

- Downstream purification of therapeutic proteins, with experience in chromatography and filtration techniques at the lab, pilot, and manufacturing scale.
- Purified therapeutic antibodies and enzymes using AKTA Avant systems with Unicorn software achieving 95-99% purity.
- Applied DOE methods to optimize buffer conditions and flow rate for maximum yield and purity at the lab-scale.
- Performed lab-scale membrane filtration and chromatography experiments to support pilot scale, then scaled up to manufacturing scale.
- Characterized purified proteins using SDS-PAGE, HPLC/MS, GC/MS, UV-Vis, and other analytical techniques to confirm purity and identity.

Education

M.S. Materials Science & Engineering — **GPA: 3.50**
Worcester Polytechnic Institute - Worcester, MA — **2024 to 2026**
Relevant coursework: Advanced Thermodynamics, Advanced Process Engineering, X-Ray Diffraction

B.S. Chemical Engineering — **GPA: 3.16**
University of Rhode Island - Kingston, RI — **2019 to 2022**
Minors: Chemistry

B.S. Applied Mathematics — **GPA: 3.16**
University of Rhode Island - Kingston, RI — **2019 to 2022**
Minors: Physics

Technical Skills

Atomistic Simulation:	DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)
Scientific Computing:	Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX, R, Unicorn
Laboratory:	Glovebox, clean-room, HPLC, AKTA FPLC, TFF, SEM, XRD, DSC/TGA, UV-Vis, Osmometry, HPLC/MS, GC/MS, pH-meter, SDS-PAGE, KF titration meter
Manufacturing:	SPC, DOE, Lean/Six Sigma, Process optimization, SOP development, Tech Transfer (MSAT), cGMP